

Fuzzy Relations

Crisp Relation

- To describe the fuzzy relation, first we describe relation by an example of daily life using discrete fuzzy sets.
- Relationship is described between the colours of a fruit X and the grade of maturity Y .
- Crisp set X with three linguistic terms is given as $X = \{\text{green, yellow, red}\}$

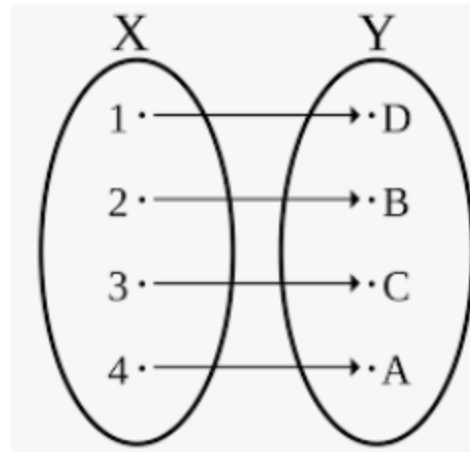
Similarly the grade of maturity for the other set Y will be

$$Y = \{\text{verdant, half-mature, mature}\}$$

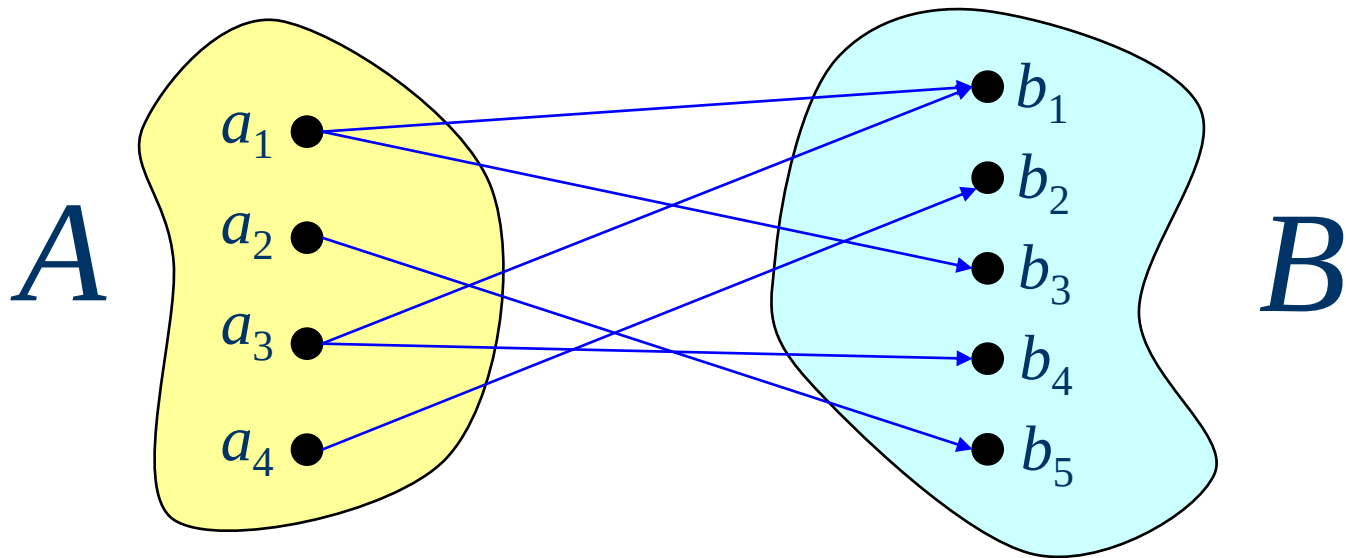
Crisp formulation of a relation $X \rightarrow Y$ between two crisp sets is presented in tabular form

	Verdant	Half-mature	Mature
Green	1	0	0
Yellow	0	1	0
Red	0	0	1

One to One Mapping

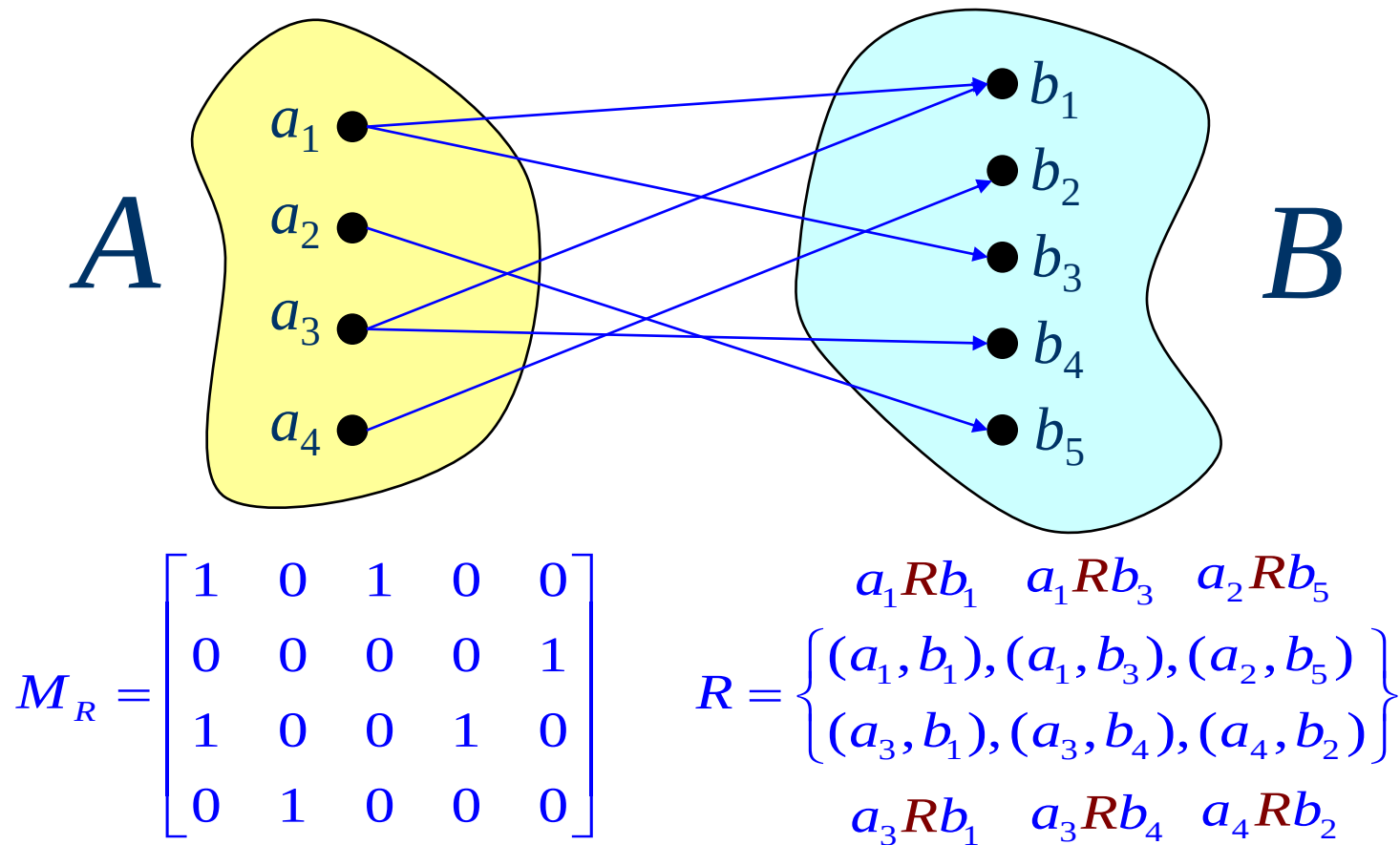


Crisp Relation (R)



$$\underline{R} \subseteq A \times B$$

Crisp Relation (R)



Crisp Relations

Example:

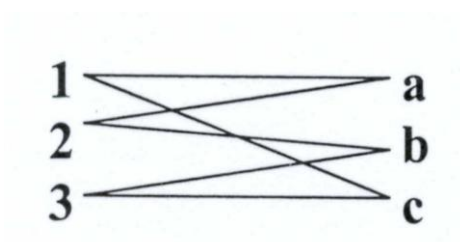
If $X = \{1,2,3\}$

$Y = \{a,b,c\}$

$R = \{ (1 a), (1 c), (2 a), (2 b), (3 b), (3 c) \}$

$$R = \begin{matrix} & \begin{matrix} a & b & c \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} \end{matrix}$$

Using a diagram to represent the relation



The Real-Life Relation

- x is close to y
 - x and y are numbers
- x depends on y
 - x and y are events
- x and y look alike
 - x and y are persons or objects
- If x is large, then y is small
 - x is an observed reading and y is a corresponding action

Fuzzy Relations

- Triples showing connection between two sets:
 $(a,b,\#)$: a is related to b with degree $\#$
- Fuzzy relations are set themselves
- Fuzzy relations can be expressed as matrices

Fuzzy Relations Matrices

- Example: Color-Ripeness relation for tomatoes

$R_1(x, y)$	unripe	semi ripe	ripe
green	1	0.5	0
yellow	0.3	1	0.4
Red	0	0.2	1

Composition

Let R be a relation that relates, or maps, elements from universe X to universe Y , and let S be a relation that relates, or maps, elements from universe Y to universe Z .

A useful question we seek to answer is whether we can find a relation, T , that relates the same elements in universe X that R contains to the same elements in universe Z that S contains. It turns out that we can find such a relation using an operation known as *composition*.

Composition

There are two common forms of the composition operation:

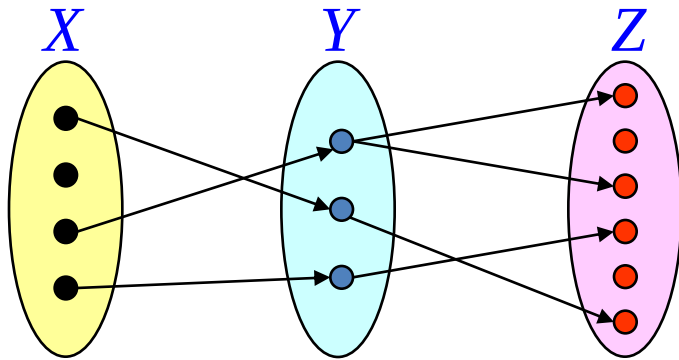
max–min composition

$$T = R \circ S,$$
$$\chi_T(x, z) = \bigvee_{y \in Y} (\chi_R(x, y) \wedge \chi_S(y, z)).$$

max–product composition.

$$T = R \circ S,$$
$$\chi_T(x, z) = \bigvee_{y \in Y} (\chi_R(x, y) \bullet \chi_S(y, z)).$$

Max-Min Composition



R : fuzzy relation defined on X and Y .

S : fuzzy relation defined on Y and Z .

$R \circ S$: the composition of R and S .
A fuzzy relation defined on X and Z .

$$\begin{aligned}\mu_{R \circ S}(x, z) &= \max_y \min(\mu_R(x, y), \mu_S(y, z)) \\ &= \bigvee_y (\mu_R(x, y) \wedge \mu_S(y, z))\end{aligned}$$

Example

$$\mu_{S \circ R}(x, y) = \max_v \min(\mu_R(x, v), \mu_S(v, y))$$

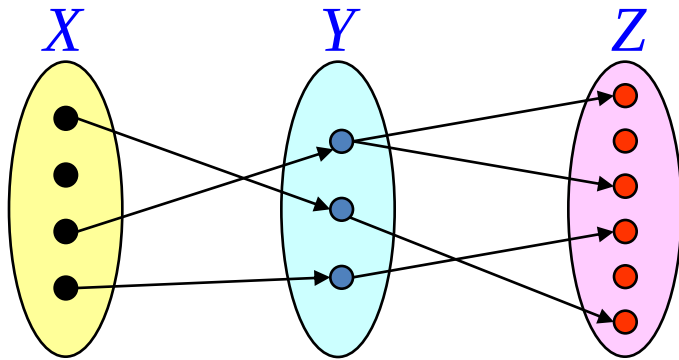
R	a	b	c	d
1	0.1	0.2	0.0	1.0
2	0.3	0.3	0.0	0.2
3	0.8	0.9	1.0	0.4

	0.1	0.2	0.0	1.0
min	0.9	0.2	0.8	0.4
max	0.1	0.2	0.0	0.4

S	α	β	γ
a	0.9	0.0	0.3
b	0.2	1.0	0.8
c	0.8	0.0	0.7
d	0.4	0.2	0.3

$R \circ S$	α	β	γ
1	0.4	0.2	0.3
2	0.3	0.3	0.3
3	0.8	0.9	0.8

Max-Product Composition



R : fuzzy relation defined on X and Y .

S : fuzzy relation defined on Y and Z .

$R \circ S$: the composition of R and S .
A fuzzy relation defined on X and Z .

$$\mu_{R \circ S}(x, y) = \max_v (\mu_R(x, v) \mu_S(v, y))$$

Example

$$\mu_{R \circ S}(x, y) = \max_v (\mu_R(x, v) \mu_S(v, y))$$

R	a	b	c	d
1	0.1	0.2	0.0	1.0
2	0.3	0.3	0.0	0.2
3	0.8	0.9	1.0	0.4

0.1 0.2 0.0 1.0

Product 0.9 0.2 0.8 0.4

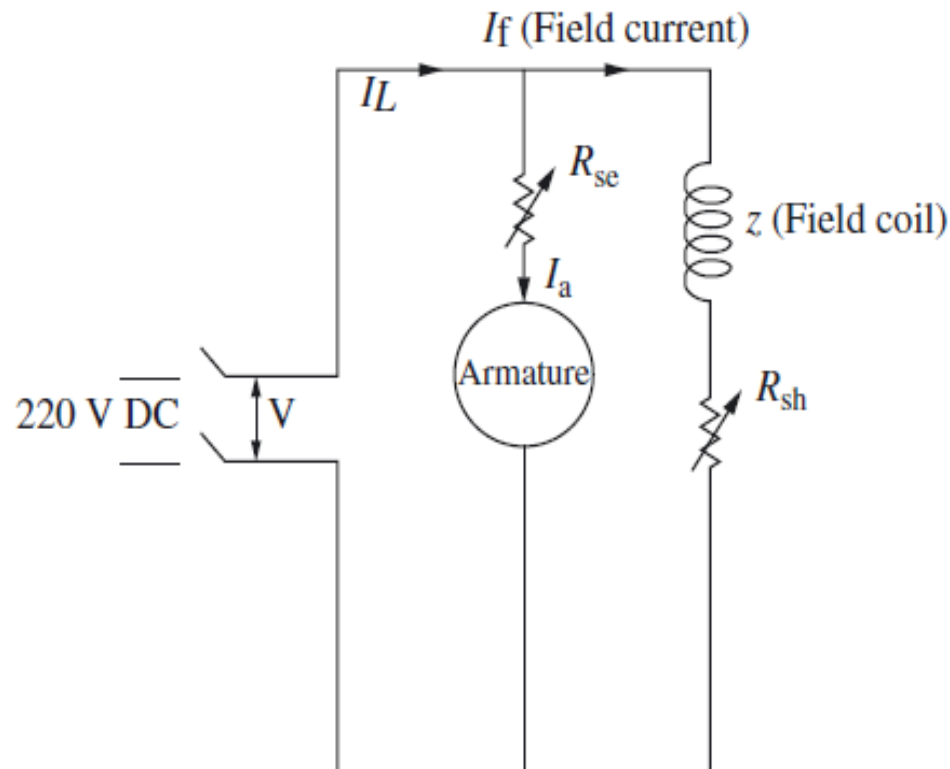
max .09 .04 0.0 0.4

S	α	β	γ
a	0.9	0.0	0.3
b	0.2	1.0	0.8
c	0.8	0.0	0.7
d	0.4	0.2	0.3

$R \circ S$	α	β	γ
1	0.4	0.2	0.3
2	0.27	0.3	0.24
3	0.8	0.9	0.7

Example

Suppose we are interested in understanding the speed control of the DC shunt motor under no-load condition, as shown.



Example

Initially, the series resistance R_{se} in should be kept in the cut-in position for the following reasons:

1. The back electromagnetic force, given by $E_b = kN\phi$, where k is a constant of proportionality, N is the motor speed, and ϕ is the flux (which is proportional to input voltage, V), is equal to zero because the motor speed is equal to zero initially.
2. We have $V = E_b + I_a(R_a + R_{se})$, therefore $I_a = (V - E_b)/(R_a + R_{se})$, where I_a is the armature current and R_a is the armature resistance. Since E_b is equal to zero initially, the armature current will be $I_a = V/(R_a + R_{se})$, which is going to be quite large initially and may destroy the armature.

Example

Let R_{se} be a fuzzy set representing a number of possible values for series resistance, say sn values, given as

$$\underline{R}_{se} = \{R_{s1}, R_{s2}, R_{s3}, \dots, R_{sn}\},$$

and let I_a be a fuzzy set having a number of possible values of the armature current, say m values, given as

$$\underline{I}_a = \{I_1, I_2, I_3, \dots, I_m\}.$$

The fuzzy sets R_{se} and I_a can be related through a fuzzy relation, say R , which would allow for the establishment of various degrees of relationship between pairs of resistance and current.

Example

Let **N** be another fuzzy set having numerous values for the motor speed, say **v** values, given as

$$\underline{N} = \{N_1, N_2, N_3, \dots, N_v\}$$

Now, we can determine another fuzzy relation, say **S**, to relate current to motor speed, that is, **Ia** to **N**.

Using the operation of composition, we could then compute a relation, say **T**, to be used to relate series resistance to motor speed, that is, **Rse** to **N**.

Example

The operations needed to develop these relations are as follows – two fuzzy Cartesian products and one composition:

$$\underset{\sim}{R} = \underset{\sim}{R}_{se} \times \underset{\sim}{I}_a,$$

$$\underset{\sim}{S} = \underset{\sim}{I}_a \times \underset{\sim}{N},$$

$$\underset{\sim}{T} = \underset{\sim}{R} \circ \underset{\sim}{S}.$$

Example

Suppose the membership functions for both series resistance **Rse** and armature current **Ia** are given in terms of percentages *of their respective rated values*, that is,

$$\mu_{R_{se}}(\%se) = \frac{0.3}{30} + \frac{0.7}{60} + \frac{1.0}{100} + \frac{0.2}{120}$$

and

$$\mu_{I_a}(\%a) = \frac{0.2}{20} + \frac{0.4}{40} + \frac{0.6}{60} + \frac{0.8}{80} + \frac{1.0}{100} + \frac{0.1}{120},$$

and the membership value for $\tilde{N}$ is given in units of motor speed in rpm,

$$\mu_{\tilde{N}}(\text{rpm}) = \frac{0.33}{500} + \frac{0.67}{1000} + \frac{1.0}{1500} + \frac{0.15}{1800}.$$

Example

The following relation then result from use of the Cartesian product to determine R:

$$\tilde{R} = \begin{array}{c} \begin{matrix} & 20 & 40 & 60 & 80 & 100 & 120 \end{matrix} \\ \begin{matrix} 30 \\ 60 \\ 100 \\ 120 \end{matrix} \left[\begin{array}{cccccc} 0.2 & 0.3 & 0.3 & 0.3 & 0.3 & 0.1 \\ 0.2 & 0.4 & 0.6 & 0.7 & 0.7 & 0.1 \\ 0.2 & 0.4 & 0.6 & 0.8 & 1 & 0.1 \\ 0.2 & 0.2 & 0.2 & 0.2 & 0.2 & 0.1 \end{array} \right] \end{array}$$

For example, $\mu_R(60, 40) = \min(0.7, 0.4) = 0.4$, $\mu_R(100, 80) = \min(1.0, 0.8) = 0.8$,

Example 3.8

Cartesian product to determine S:

$$\tilde{S} = \begin{array}{c} 20 \\ 40 \\ 60 \\ 80 \\ 100 \\ 120 \end{array} \begin{array}{cc} 500 & 1000 & 1500 & 1800 \\ \left[\begin{array}{cccc} 0.2 & 0.2 & 0.2 & 0.15 \\ 0.33 & 0.4 & 0.4 & 0.15 \\ 0.33 & 0.6 & 0.6 & 0.15 \\ 0.33 & 0.67 & 0.8 & 0.15 \\ 0.33 & 0.67 & 1 & 0.15 \\ 0.1 & 0.1 & 0.1 & 0.1 \end{array} \right] \end{array}.$$

$$\mu_{\tilde{S}}(80, 1000) = \min(0.8, 0.67) = 0.67.$$

Example

The following relation results from a max–min composition for T:

$$\mathbb{T} = \mathbb{R} \circ \mathbb{S} = \begin{array}{c} 30 \\ 60 \\ 100 \\ 120 \end{array} \begin{array}{c} 500 \quad 1000 \quad 1500 \quad 1800 \\ \left[\begin{array}{cccc} 0.3 & 0.3 & 0.3 & 0.15 \\ 0.33 & 0.67 & 0.7 & 0.15 \\ 0.33 & 0.67 & 1 & 0.15 \\ 0.2 & 0.2 & 0.2 & 0.15 \end{array} \right] \end{array}.$$

$$\begin{aligned} \mu_{\mathbb{T}}(60, 1500) &= \max[\min(0.2, 0.2), \min(0.4, 0.4), \min(0.6, 0.6), \\ &\quad \min(0.7, 0.8), \min(0.7, 1.0), \min(0.1, 0.1)]. \\ &= \max[0.2, 0.4, 0.6, 0.7, 0.7, 0.1] = 0.7. \end{aligned}$$

HW

Using max-min find the relational matrix between R_1 and R_2

$$R_1 = \begin{matrix} & \begin{matrix} y_1 & y_2 & y_3 & y_4 & y_5 \end{matrix} \\ \begin{matrix} x_1 \\ x_2 \\ x_3 \end{matrix} & \begin{bmatrix} 0.1 & 0.2 & 0.1 & 1 & 0.8 \\ 0.4 & 0.5 & 0 & 0.2 & 1 \\ 0.9 & 0.2 & 0.4 & 0.3 & 0.2 \end{bmatrix} \end{matrix},$$

$$R_2 = \begin{matrix} & \begin{matrix} z_1 & z_2 & z_3 & z_4 \end{matrix} \\ \begin{matrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \end{matrix} & \begin{bmatrix} 0.8 & 0.1 & 0.5 & 0.4 \\ 0.3 & 0.9 & 0.8 & 0.1 \\ 1 & 0.2 & 0.6 & 0.1 \\ 0.4 & 0.2 & 0.3 & 0 \\ 0.1 & 1 & 0.8 & 0.7 \end{bmatrix} \end{matrix}.$$